Chris Cameron

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AI-native researcher, developer, and mechatronics engineer completing a PhD in electrophysiology (expected early 2027). I also lecture part-time in mechatronics engineering, teaching computer architecture, C programming, circuit design, and embedded systems.

Cubie — GPU numerical integration

github.com/cubiepy/cubie

I built Cubie, an open-source Python ODE solver for consumer NVIDIA GPUs, to support electrophysiology simulations for a real-time application.

Current benchmarks (draft, unpublished) show **10,000× speedup** over CPU implementations (SciPy, MATLAB) and **100×** over the nearest GPU competitor, DiffEqGPU.jl, on real-world problems.
Benchmarks: github.com/ccam80/GPUODEBenchmarks

After adapting the algorithms for CUDA and addressing tiling and memory placement, I improved performance by:

- using NVIDIA profiling tools to trace bottlenecks and stalls to individual instructions;
- using measured instruction counts to choose loop unrolling that fits the instruction cache;
- reducing occupancy to lower L2 cache misses;
- reordering and fusing operations across each kernel to shorten live ranges and reduce register spilling; and
- placing buffers in shared or local memory based on working-set size, trading guaranteed on-chip storage against reliance on caches.

Cubie's numerical correctness is critical to its use as a scientific tool. To maintain correctness as the codebase evolves, I:

- test numerical results against known references and custom CPU harnesses;
- review every pull request line by line, checking for errors in AI-generated code; and
- require automated correctness checks and a ratcheting performance benchmark before approving each pull request.

Open-source contributions

numba-cuda / numba-cuda-mlir

My merged and open pull requests address compiler performance, typing and lowering correctness, AST transforms, and kernel performance controls.

SymPy and lcapy — minor edits.